

Ensemble Models in Natural Language Processing

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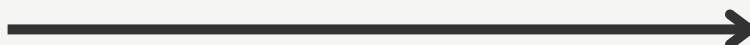
Background



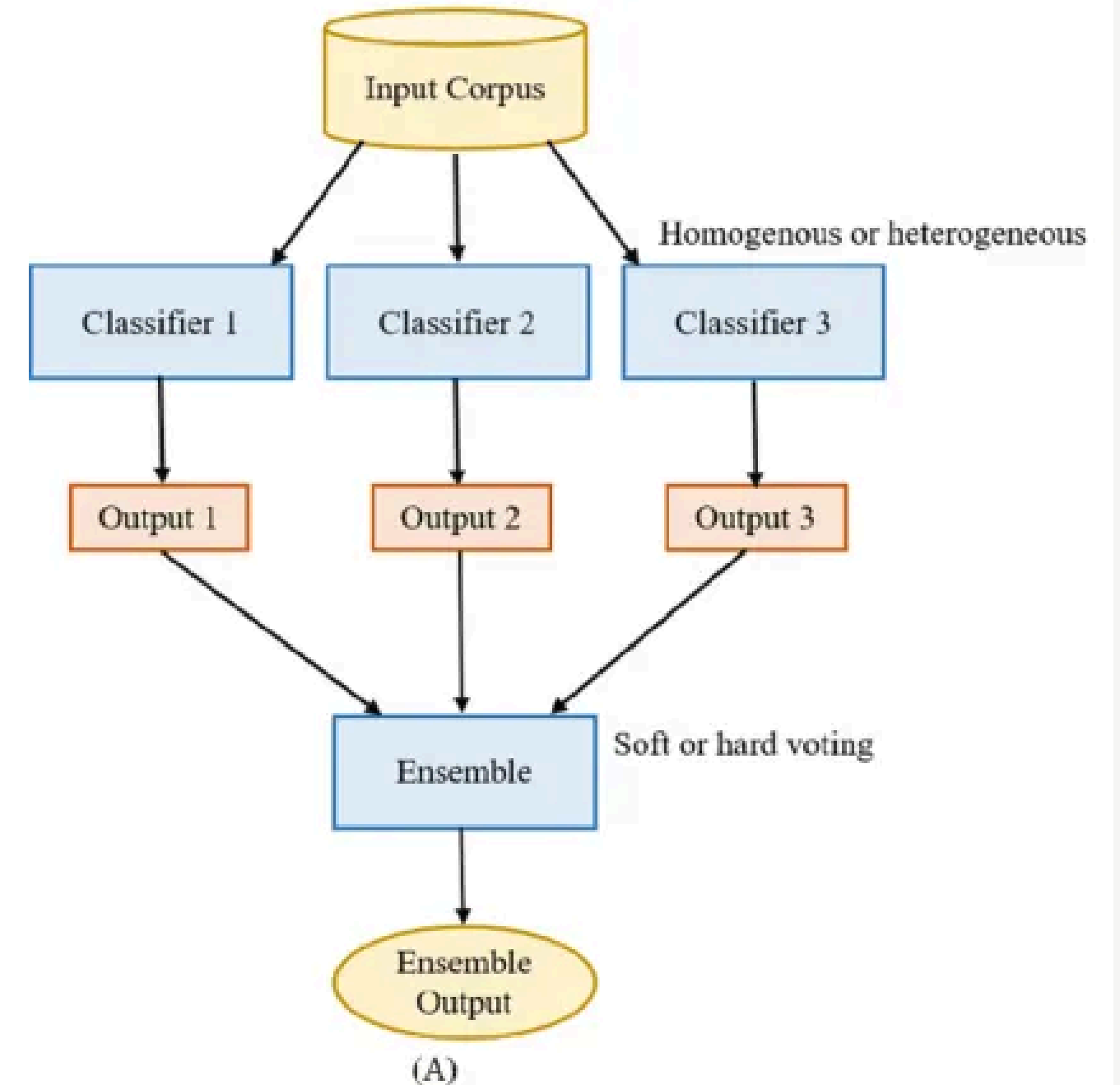
- Large Language Models show remarkable performance across diverse tasks.
- However, they have their own distinct strength and weaknesses in different tasks due to variation in:
 - architectures,
 - model sizes,
 - training data[2]
- Single LLMs face several challenges, some of them are:
 - susceptibility to biases,
 - high computational requirements, and
 - issues related to interpretability[1]



What are ensemble models?



- A combination of multiple models known as base or weak learners to achieve superior predictive performance. [1]
- In the context of NLP, the goal is to aggregate the distinct strengths of different LLMs to achieve consistent performance across diverse tasks. [2]
- The underlying principle is to exploit the diversity of predictions to reduce errors, enhance robustness, and improve generalization. [1]



Types of Ensemble Models



[1]

Ensemble Type	Typical Examples	Advantages	Limitations
Model-level	Bagging, boosting, stacking, voting	High predictive performance, robust generalization, reduces variance	High computational cost, high latency, low interpretability
Parameter-level	Snapshot ensembles, parameter averaging	Computational efficiency, improved scalability, moderate performance improvement	Limited model diversity, lower interpretability, moderate accuracy gains
Task-specific	Prompt-based ensembles, domain-specific fine-tuned ensembles	High domain specialization, scalable, adaptable to targeted applications	Dependent on prompt and domain quality, less generalizable across domains
Knowledge distillation	Student-teacher model training, distilled compact LLMs	High computational efficiency, lower inference latency, suitable for deployment in resource-constrained environments	Moderate performance reduction, requires careful distillation tuning
Mixture-of-experts	Switch transformers, sparsely gated experts	Exceptional scalability, efficient resource utilization, capable of training very large models	Complex gating mechanisms, interpretability issues, expert load balancing
Hybrid/multi-agent	Collaborative agent frameworks, generative agent ensembles	Effective in complex, multi-step reasoning tasks, interactive capabilities, adaptable in dynamic environments	High coordination complexity, synchronization overhead, limited interpretability
Multimodal LLM ensembles	Late fusion, modality-specific ensembling, alignment-based fusion	Cross-modal reasoning, robust in heterogeneous data settings, excels in VQA and clinical support tasks	Modality alignment, synchronization, and scalability challenges

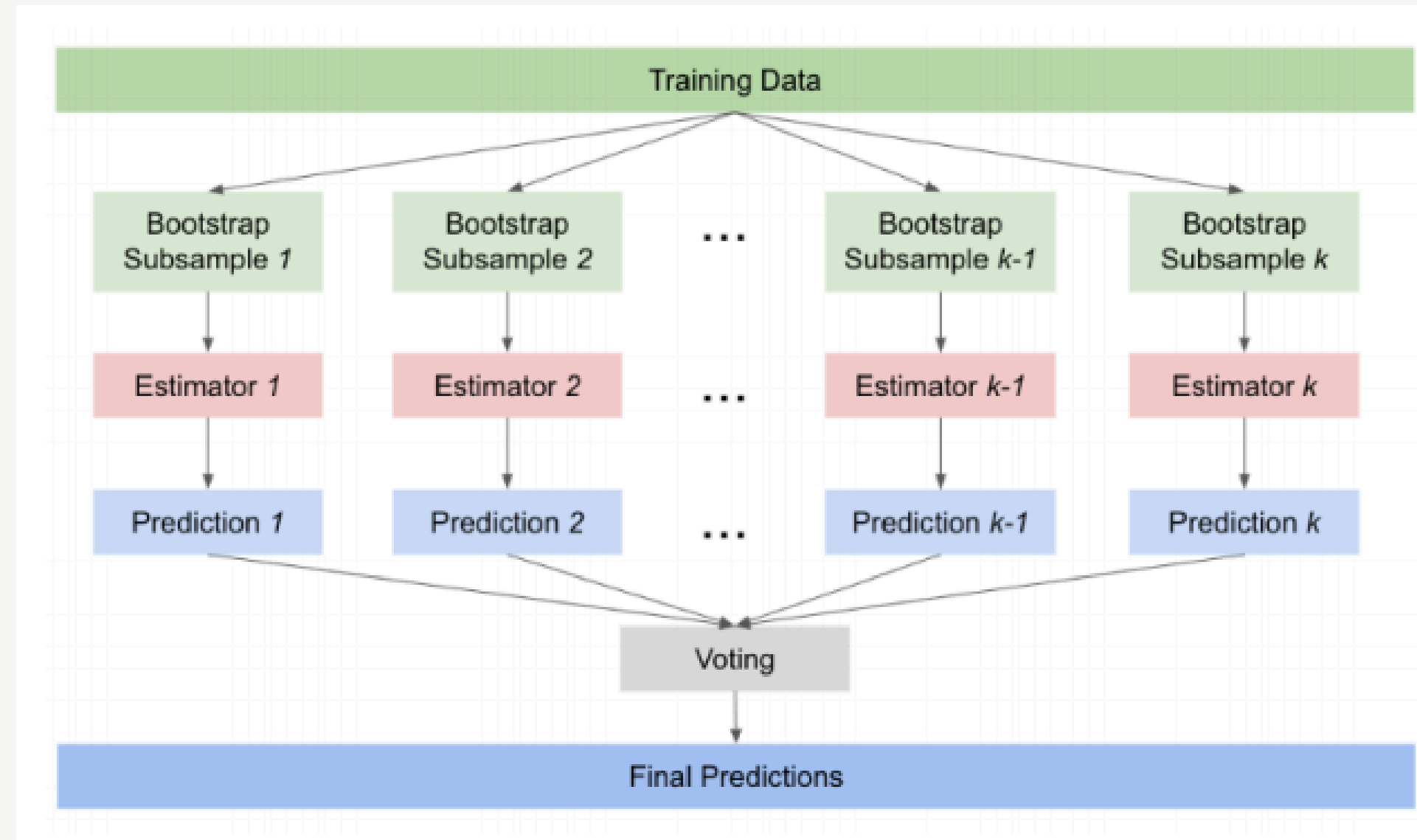
Model Level Ensembles

Bagging (Bootstrap Aggregating)

- Trains base models on different bootstrap samples of the training data
- The final ensemble prediction is a weighted sum of individual models' outputs:

$$H(x) = \frac{1}{n} \sum_{i=1}^n h_i(x),$$

- Reduces overfitting and variance, often resulting in more stable predictions[1]



Model Level Ensembles

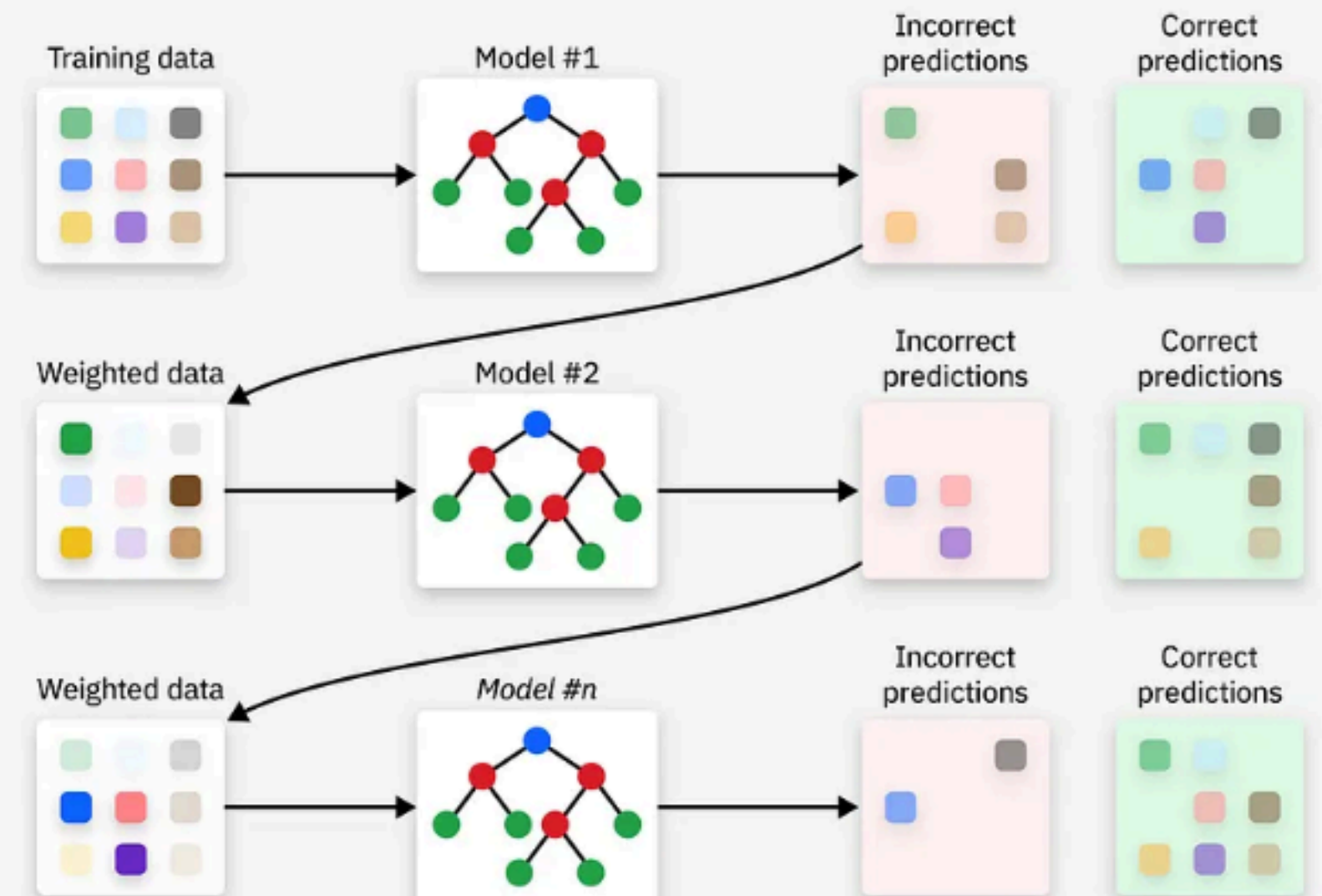
Boosting

- Sequentially trains models, with each model focusing on correcting the errors made by its predecessor.
- The final prediction is a weighted sum of individual models' outputs:

$$H(x) = \sum_{i=1}^n w_i h_i(x),$$

where w_i represents the weight of the i -th model, often proportional to its accuracy

- Gives higher weight to the training examples that previous models misclassified.[1]
- Effective for tasks that demand high precision, such as NER or text classification



[1]

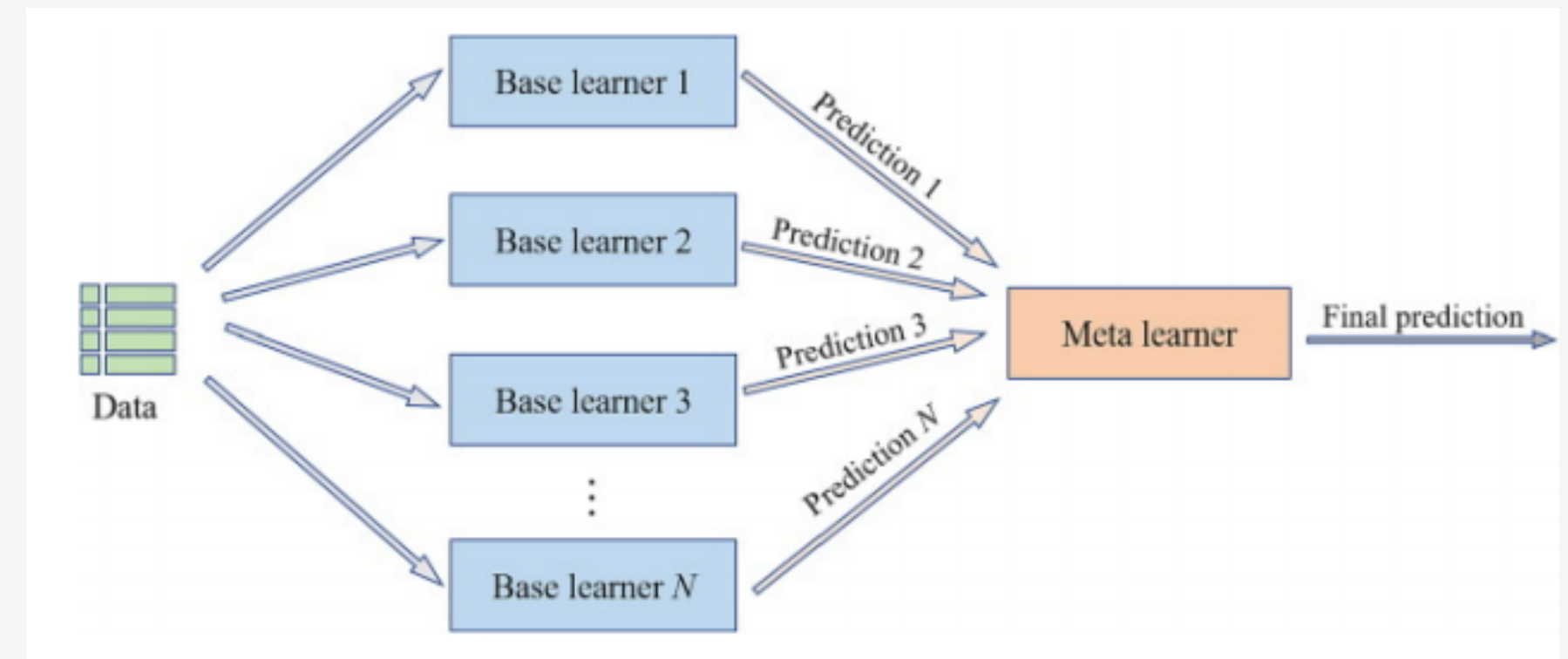
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Model Level Ensembles

Stacking (Stacked Generalization)

[1]

- Combines different prediction models in layers using a meta-model.
- The meta-model is trained using the outputs of the base models as features, enabling it to learn dependencies among them. [1]
- For example, this is how meta-model makes predictions using base models A,B,C:
 - If A says "negative" but B says "positive" and review length > 200 → trust B
 - If C says "positive" but rare-word frequency is high → reduce confidence.
 - If B is uncertain and A & C agree → follow A and C
- Aims to minimize the errors of generalization by reducing the bias of its generalizers [3]



Meta Learner input Matrix:

$$Z = \begin{bmatrix} \hat{y}_1^{(1)} & \hat{y}_1^{(2)} & \dots & \hat{y}_1^{(M)} \\ \hat{y}_2^{(1)} & \hat{y}_2^{(2)} & \dots & \hat{y}_2^{(M)} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{y}_N^{(1)} & \hat{y}_N^{(2)} & \dots & \hat{y}_N^{(M)} \end{bmatrix}$$

Target matrix:

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}$$

M models and N validation examples

Model Level Ensembles

Voting Ensembles

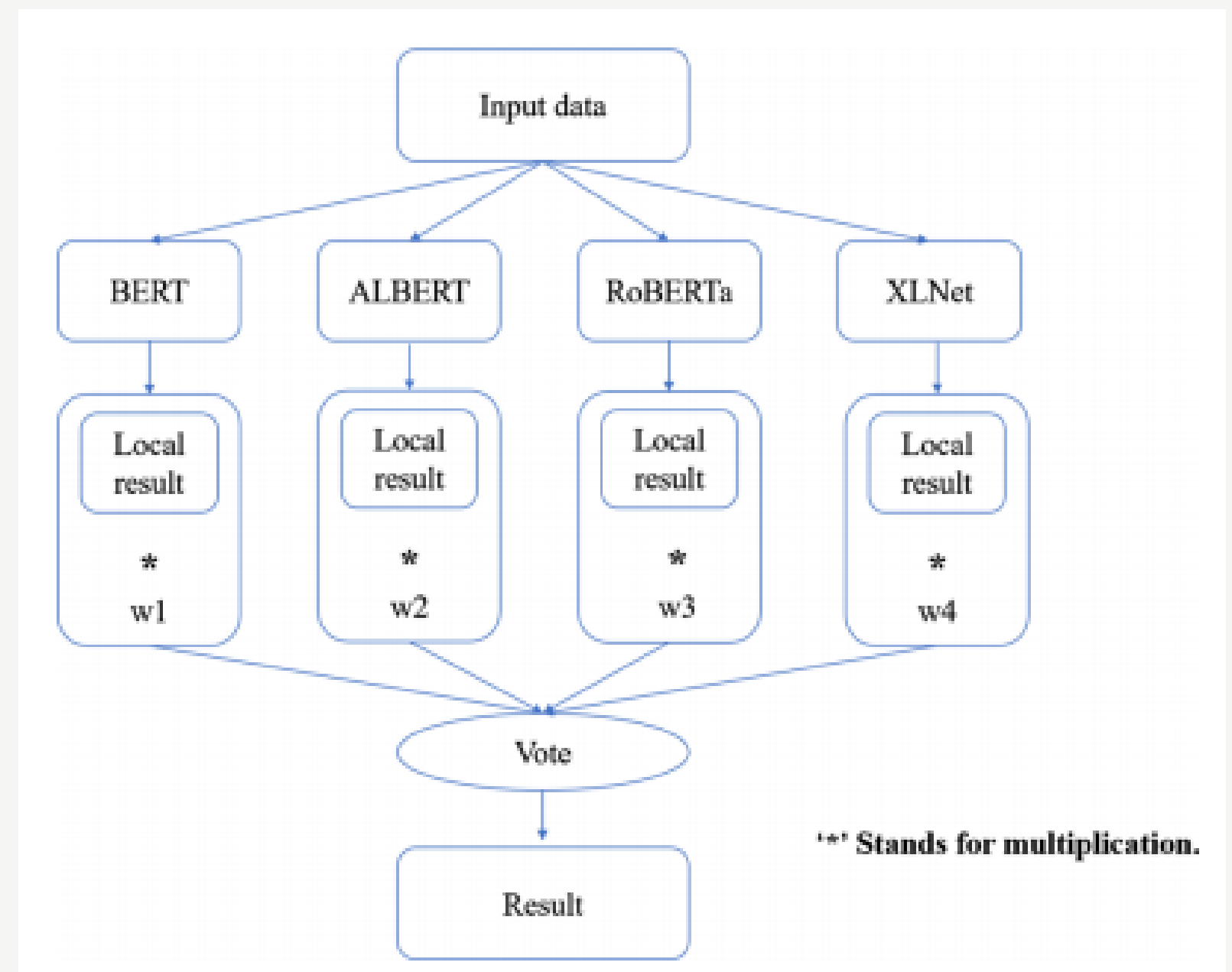
[1]

- Combines predictions from multiple models using weighted voting. For classification tasks, the ensemble prediction is:

$$H(x) = \arg \max_k \sum_{i=1}^n w_i \cdot \mathbb{I}\{h_i(x) = k\},$$

Dry Run Example:

- Models:** h1, h2, h3, **Weights:** w1=0.5, w2=0.3, w3=0.2
- Predictions for input x:
- h1(x)=A, h2(x)=B, h3(x)=A
- For class A: $\sum w_i \cdot \mathbb{I}\{h_i(x)=A\} = (0.5 \cdot 1) + (0.3 \cdot 0) + (0.2 \cdot 1) = 0.7$
- For class B: $(0.5 \cdot 0) + (0.3 \cdot 1) + (0.2 \cdot 0) = 0.3$
- For class C: $(0.5 \cdot 0) + (0.3 \cdot 0) + (0.2 \cdot 0) = 0$
- Useful in QA systems, where models fine-tuned on diverse datasets independently generate answers. [1]



Mixture of Expert Methods

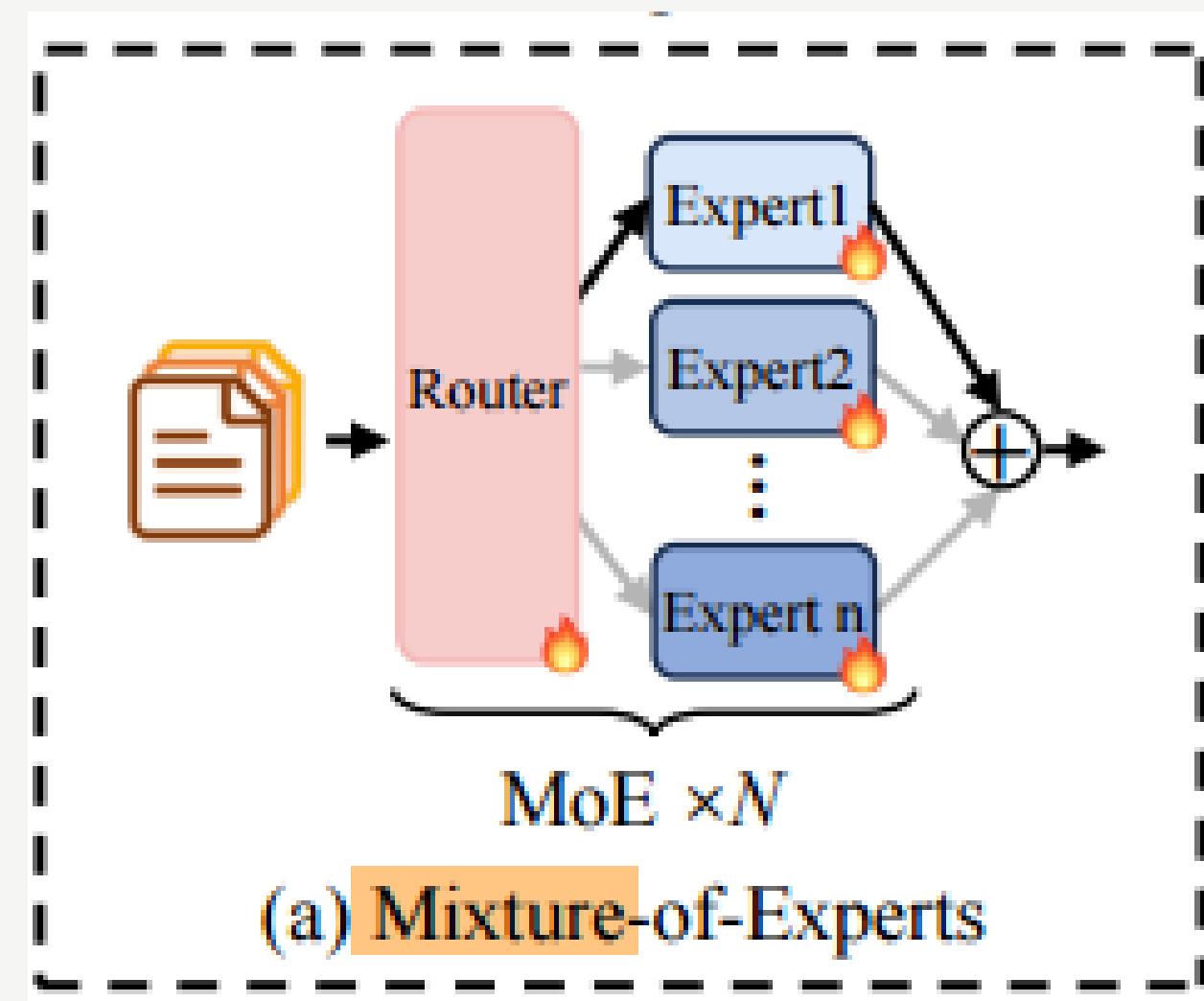


- These methods use a router network to select and activate only a subset of experts to process a given input, thereby aggregating expertise [2]

$$y = \sum_{i=1}^n G_i(x) \cdot E_i(x),$$

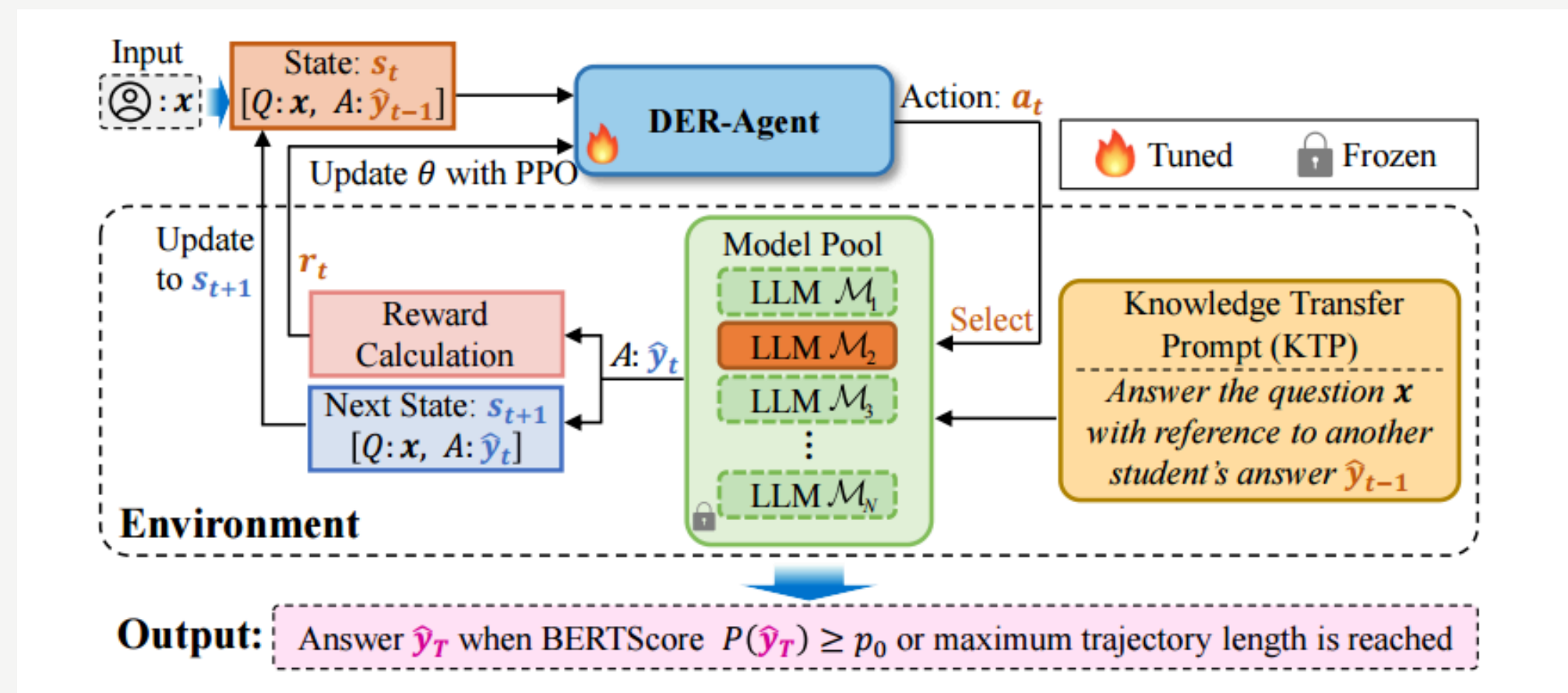
- where $G_i(x)$ represents the gating weight for the i -th expert, satisfying the condition $\sum_{i=1}^n G_i(x) = 1$.
- Instead of one model handling everything, MoE divides the work:
 - Expert 1 → maybe good at mathematical reasoning
 - Expert 2 → maybe good at dialogue
 - Expert 3 → maybe good at summarization
- In many modern MoE systems (e.g., Switch Transformer) the gate selects one expert (top-1 routing), which is ultra efficient [5]

[5]



Dynamic Ensemble Reasoning (DER)

- Views the LLM ensemble problem as a sequential decision-making challenge and models it as a Markov Decision Process (MDP)
- The agent takes the **state (S)** as input, which consists of the question and the answer from the previous step: $s_t = [Q: x, A: \hat{y}_{t-1}]$.
- The agent performs an **action (A)** by selecting the next LLM from the model pool

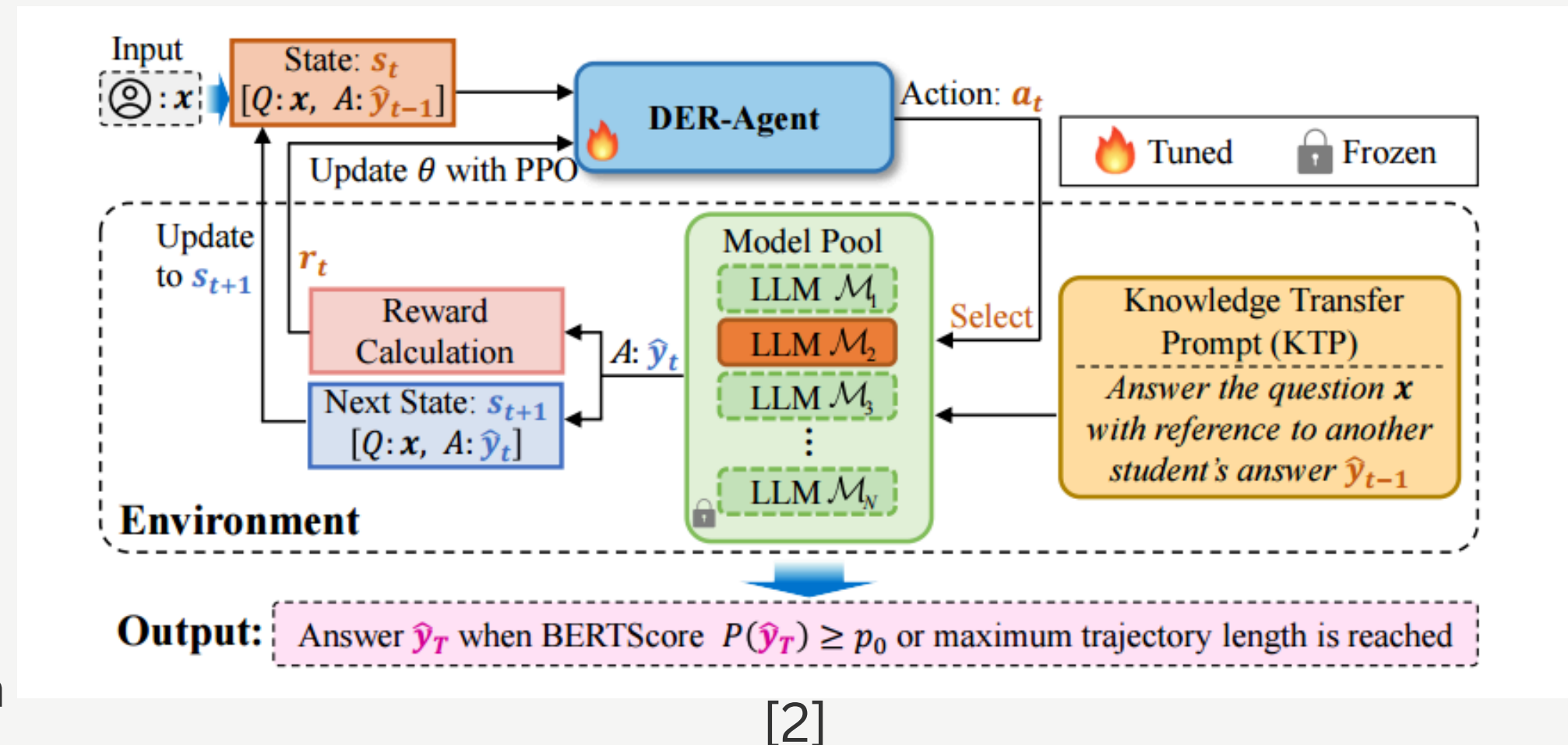


[2]

- The selected LLM then generates a new answer ($\hat{y}_t$), updating the state to s_{t+1} . **Transition T (S, A)** is a function which maps a state s_t into a new state s_{t+1} .
- Reward(R)** can be considered as an evaluation of the quality of the answer $\hat{y}$ for the selected LLM.

Dynamic Ensemble Reasoning (DER)

- **The Knowledge Transfer Prompt (KTP)** instructs the subsequent LLM to refer to the previous answer ($\hat{y}_{t-1}$) as an "another student's answer".
- Promotes the current LLM to generate a more satisfactory answer and avoids the overriding influence of the predecessor's answer
- **Reward Function (R_t)**: The reward function is designed to optimize the route selection based on three core aspects:
 - **Answer Quality**: $P(\hat{y}_t)$ (using BERTScore).
 - **Increment Quality**: $\Delta P(\hat{y}_t) = P(\hat{y}_t) - P(\hat{y}_{t-1})$ (the performance gain from the previous step).
 - **Computational Cost**: A penalty for the computation cost $C(\text{Mat})$ of the selected LLM



Dynamic Ensemble Reasoning (DER)

Category	Methods	Param. ↓ Agent Infer.		s/sample ↓	Rouge-S ↑	BARTScore ↑	BLEURT ↑	BERTScore ↑	GPT-Rank ↓
LLMs	Oracle (BARTScore)	-	-	-	0.33	-2.87	-0.38	73.23	-
	ChatGPT [Achiam <i>et al.</i> , 2023]	-	≈175B	-	0.39	-3.00	-0.26	76.23	-
	MOSS [Sun <i>et al.</i> , 2023]	-	16B	-	0.19	-3.69	-0.73	64.85	-
	Vicuna [Chiang <i>et al.</i> , 2023]	-	13B	-	0.27	-3.44	-0.61	69.60	-
	Alpaca [Taori <i>et al.</i> , 2023]	-	13B	-	0.29	-3.57	-0.53	71.46	-
	Baize [Xu <i>et al.</i> , 2023]	-	13B	-	0.20	-3.53	-0.66	65.57	-
	Open Assistant [LAION-AI, 2023]	-	12B	-	0.34	-3.45	-0.39	74.68	-
	Dolly2 [Conover <i>et al.</i> , 2023]	-	12B	-	0.16	-3.83	-0.87	62.26	-
	FLAN-T5 [Chung <i>et al.</i> , 2024]	-	11B	-	0.13	-4.57	-1.23	64.92	-
	Koala [Geng <i>et al.</i> , 2023]	-	7B	-	0.19	-3.85	-0.84	63.96	-
	Mosaic MPT [Team and others, 2023]	-	7B	-	0.14	-3.72	-0.82	63.21	-
	StableLM [Stability-AI, 2023]	-	7B	-	0.17	-4.12	-0.98	62.47	-
	ChatGLM [Du <i>et al.</i> , 2022]	-	6B	-	0.27	-3.52	-0.62	70.38	-
Ensemble	Classifier-OPT	125M	13B	3.98	0.27	-3.44	-0.61	69.60	2.84
	PairRanker [Jiang <i>et al.</i> , 2023]	400M	117B	44.41	0.32	-3.14	-0.38	73.03	2.25
	LLM-debate [Du <i>et al.</i> , 2023]	-	234B	56.58	0.27	-3.51	-0.55	71.59	3.16
	ReConcile [Chen <i>et al.</i> , 2024]	-	351B	55.75	0.24	-3.79	-0.71	68.61	3.68
	sampling-voting [junyou li <i>et al.</i> , 2024]	110M	234B	38.07	0.27	-3.39	-0.33	70.12	3.20
	DyLAN [Liu <i>et al.</i> , 2024]	-	>234B	115.68	0.28	-3.69	-0.64	70.90	3.85
DER (Ours)		125M	17B	9.32	0.35(+9.38%)	-3.14(+0.00%)	-0.31(+6.06%)	75.00(+2.70%)	2.02

Table 1: Comparison with LLMs and state-of-the-art baselines on the *MixInstruct*. (·) indicates relative improvement over the second-best.

QA task

- DER is consistently better than single LLM for QA task.
- Crucially, the DER on BARTScore is improved by 8.7% compared to Vicuna.
- DER reaches 98% of the ChatGPT performance on the BERTScore, while the inference parameters are only 10% of those of ChatGPT

Dynamic Ensemble Reasoning (DER)

Category	Methods	Param. ↓		Acc (%) ↑	Rouge-S ↑	BARTScore ↑	BLEURT ↑	BERTScore ↑	GPT-Rank ↓
		Agent	Infer.						
LLMs	MOSS [Sun <i>et al.</i> , 2023]	–	16B	19.41	0.29	-5.27	-0.58	70.14	-
	Vicuna [Chiang <i>et al.</i> , 2023]	–	13B	33.74	0.36	-4.51	-0.43	74.68	-
	Alpaca [Taori <i>et al.</i> , 2023]	–	13B	8.11	0.17	-5.86	-0.95	63.07	-
	Baize [Xu <i>et al.</i> , 2023]	–	13B	31.08	0.30	-5.11	-0.59	69.97	-
	Open Assistant [LAION-AI, 2023]	–	12B	16.60	0.31	-5.26	-0.58	70.64	-
	Dolly2 [Conover <i>et al.</i> , 2023]	–	12B	11.52	0.23	-5.66	-0.88	67.27	-
	FLAN-T5 [Chung <i>et al.</i> , 2024]	–	11B	24.03	0.37	-4.94	-0.42	74.68	-
	Koala [Geng <i>et al.</i> , 2023]	–	7B	20.70	0.25	-4.94	-0.95	67.98	-
	Mosaic MPT [Team and others, 2023]	–	7B	23.43	0.31	-4.97	-0.53	70.77	-
	StableLM [Stability-AI, 2023]	–	7B	19.56	0.32	-4.70	-0.39	74.76	-
Ensembles	ChatGLM [Du <i>et al.</i> , 2022]	–	6B	21.08	0.33	-4.90	-0.57	72.64	-
	PairRanker [Jiang <i>et al.</i> , 2023]	400M	117B	30.17	0.36	-4.74	-0.39	74.26	1.85
	LLM-debate [Du <i>et al.</i> , 2023]	-	234B	19.71	0.27	-5.35	-0.74	68.39	2.43
	Ensemblesampling-voting [junyou li <i>et al.</i> , 2024]	110M	234B	26.08	0.36	-4.91	-0.34	73.21	1.68
	DyLAN [Liu <i>et al.</i> , 2024]	-	>234B	19.64	0.25	-5.41	-0.77	68.02	2.77
DER (Ours)		125M	26B	34.98_(+15.94%)	0.37	-4.44	-0.41	75.14	1.27

Table 2: Comparison with LLMs and state-of-the-art baselines on the *GSM8K*.

- DER outperformed state-of-the-art methods on the GSM8K mathematical reasoning task, improving accuracy by about 16% compared to PairRanker

Mathematical Reasoning task

Applications *For NLP*

Question Answering: Ensembles are crucial for improving the accuracy and reasoning in QA systems[4]. They have been successfully applied to open-domain question answering [5] and specialized medical QA systems[1]

Sentiment Analysis and Text Classification: Ensembles combine diverse models (like BERT, XLNet, RoBERTa, ALBERT, GPT2) to improve accuracy and F1 scores in sentiment analysis. [1] [6] They are also used for classifying Chinese medical texts. [1]

Named Entity Recognition (NER): Ensembles enhance the precision of NER and relation extraction tasks, particularly in complex fields like biomedical information extraction[1] [6]

Text Summarization: Ensembling techniques, such as those combining outputs from multiple LLMs at the character level, are used to enhance accuracy in text generation. [5] They are also explored for summarization tasks. [1]

Code Generation and Mathematical Problem-Solving: MoE models (e.g., Mixtral-8x7B, DeepSeekMoE) are instrumental in code generation and complex mathematical problem-solving tasks[1],[5]

Applications

→

Domain Specific

Application Domain	Reference	Year	Description
Model robustness enhancement	Cohen et al. [80]	2025	Introduced DFPE, a fingerprint-based ensemble for performance and diversity.
Software fault localization	Cho et al. [82]	2025	Used ensembles of small LLMs to improve bug and fault detection in codebases.
Medical question answering	Lucas et al. [69]	2024	Applied LLM ensembles for improved reasoning in medical QA tasks.
Biomedical information extraction	Li et al. [70]	2024	Used an ensemble of pretrained LLMs to extract biomedical entities and relations from literature.
EHR annotation	Huang et al. [37]	2024	Introduced a multi-agent LLM ensemble to streamline annotation of electronic health records.
Health data imputation	He et al. [68]	2024	Proposed LLM-Forest to enhance imputation of missing values in structured health datasets.
Open healthcare NLP	Gururajan et al. [72]	2024	Released Aloe, a family of fine-tuned open healthcare LLMs for ensemble clinical NLP.
LLM output detection	Lai et al. [73]	2024	Deployed adaptive transformer ensembles to detect AI-generated clinical text.
Paper provenance tracing	Chen et al. [83]	2024	Proposed a GPU-free ensemble LLM method to trace origins of academic publications.
General NLP tasks	Huang et al. [67]	2024	Introduced DEEPEN, a framework for combining heterogeneous LLMs with parallel token alignment.
LLM diversity optimization	Tekin et al. [81]	2024	Proposed LLM-TOPLA, which maximizes diversity among LLMs for more robust NLP outputs.
Vocabulary alignment	Xu et al. [11]	2024	Presented techniques to harmonize vocabularies across ensemble LLMs for consistency.
Cross-lingual sentiment analysis	Miah et al. [77]	2024	Created a multimodal LLM ensemble for sentiment analysis across multiple languages.
Data annotation	Farr et al. [38]	2024	Proposed LLM chain ensembles to improve annotation efficiency and scale.

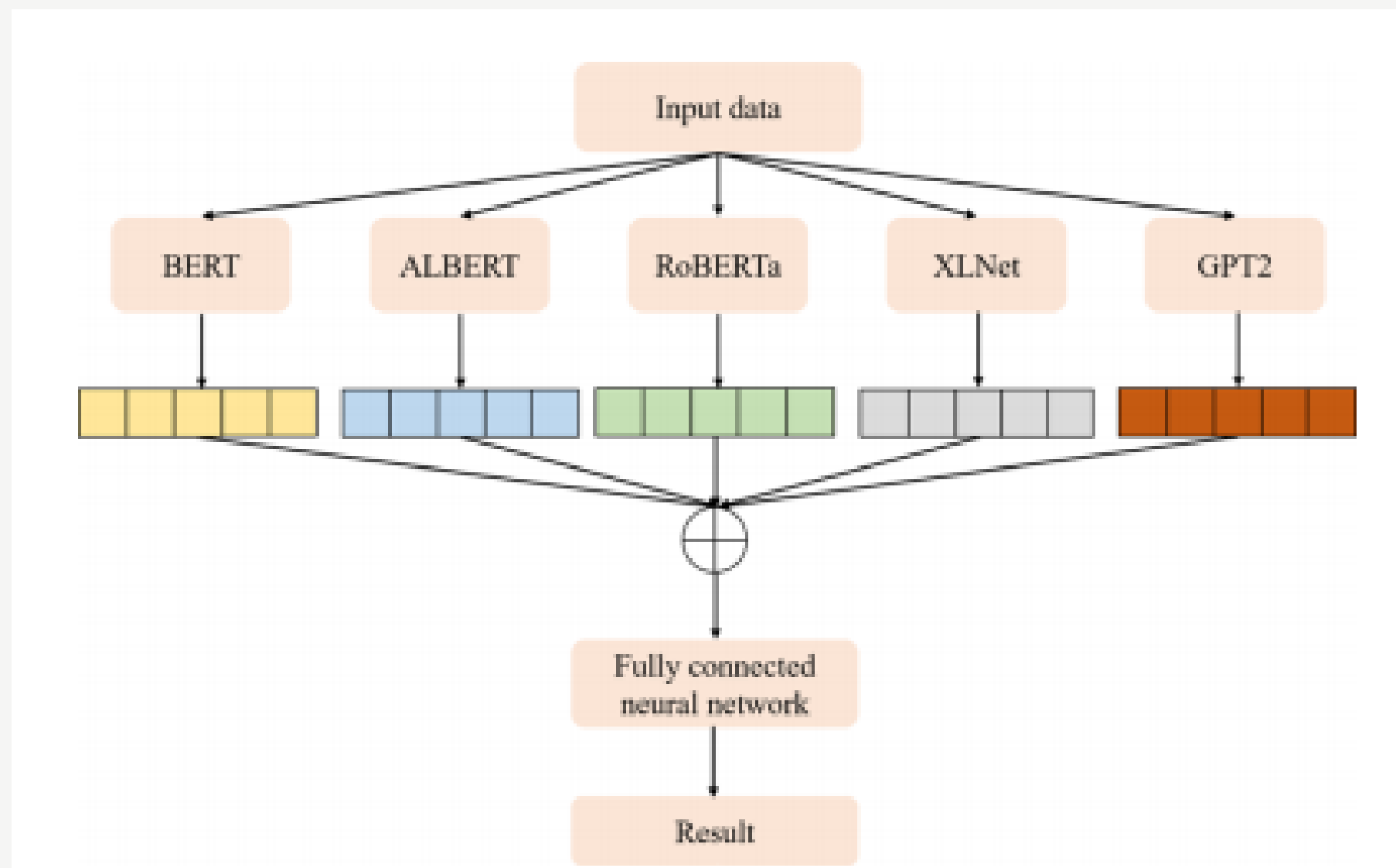
Query reformulation	Dhole and Agichtein [76]	2024	Introduced GenQRensemble for zero-shot query rewriting using LLM ensembles.
Product attribute extraction	Fang et al. [12]	2024	Built an ensemble model to extract attributes from e-commerce product data.
Smart contract analysis	Luo et al. [78]	2024	Proposed FELLMVP, an LLM ensemble to detect vulnerabilities in smart contracts.
Composed image retrieval	Yang et al. [85]	2024	Used LLM ensembles for zero-shot composed image retrieval via divergent reasoning.
Text recognition	Huang [79]	2024	Combined BERT and BPE-based LLMs for improved optical text recognition.
Code generation and math reasoning	Gu et al. [75]	2024	Proposed CharED, a character-wise ensemble decoding approach enhancing LLM performance in programming and mathematical tasks.
Medical question answering	Yang et al. [33]	2023	Developed an ensemble pipeline combining LLMs to improve performance on medical QA benchmarks.
Medical text classification	Wu et al. [71]	2023	Implemented LLM ensembles to classify Chinese medical texts for healthcare informatics.

Performance $\longrightarrow$ *Sentiment Analysis*

- The five transformer models are trained separately, and then the model is evaluated through the accuracy of the model and the confusion matrix
- The outputs are cascaded and then put into a deep neural network for training. $\rightarrow$ Stacking
- The ensemble lg model we proposed can shows best accuracy and F1 score

Table 16 Results for sentiment analysis task

Model	BERT	GPT2	XLNet	RoBERTa	ALBERT	Ensemble learning
Acc	0.764	0.647	0.688	0.717	0.766	0.787
F1	0.774	0.660	0.703	0.729	0.775	0.795
P	0.779	0.676	0.704	0.733	0.788	0.811
R	0.773	0.652	0.706	0.728	0.767	0.786
AUC	0.942	0.859	0.918	0.927	0.935	0.953



[6]

Performance $\longrightarrow$ NER

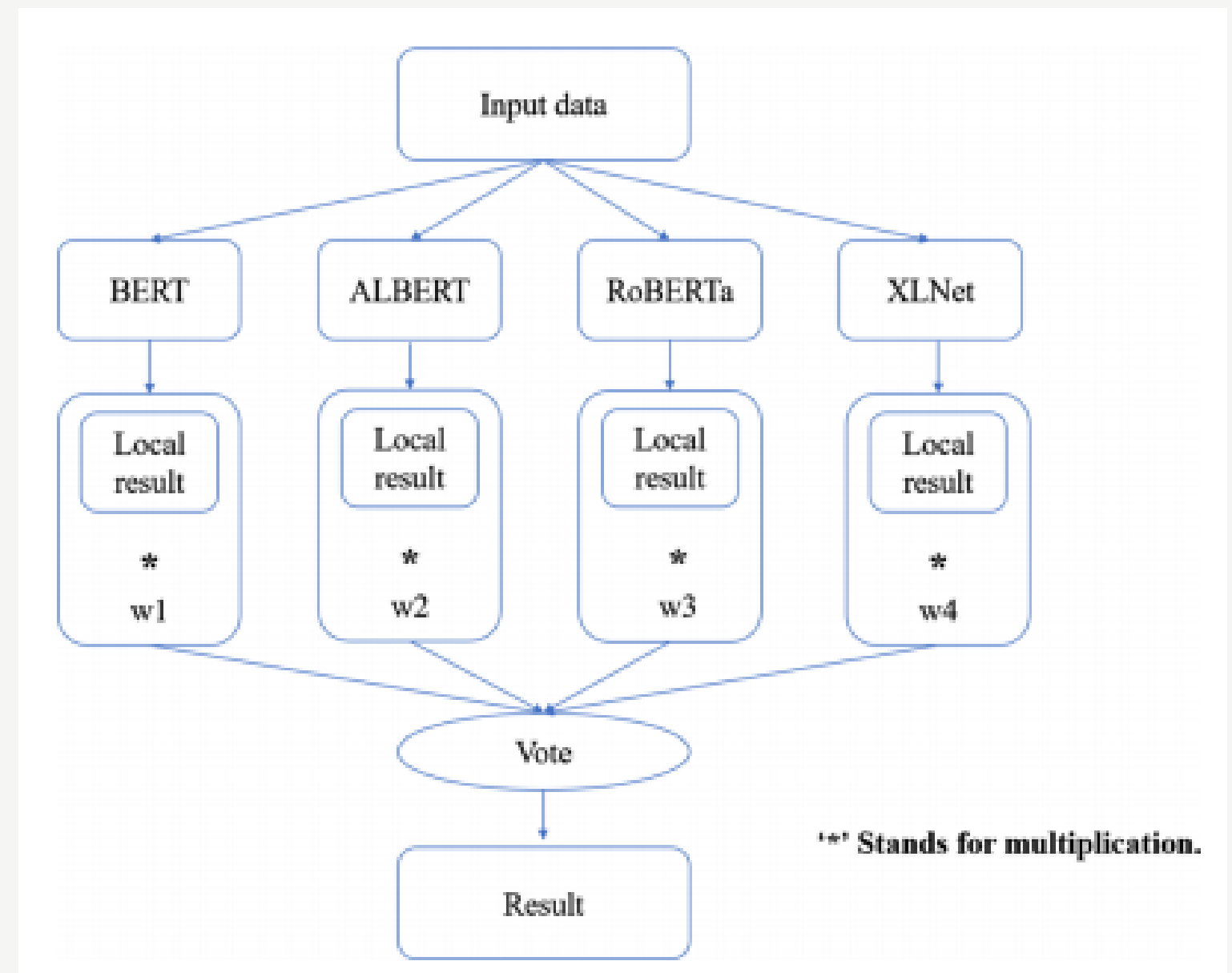
- There are four single classifiers, so there are four sets of prediction results
- The accuracy rate is used as the weight of the of a single model.
- The ensemble proposed shows best accuracy.

Table 17 The weight of different classifiers

Model	w1	w2	w3	w4
Weight	0.938	0.951	0.894	0.964

Table 18 Results for NER task

Model	BERT	XLNet	RoBERTa	ALBERT	Ensemble learning
Acc	0.938	0.951	0.886	0.894	0.964



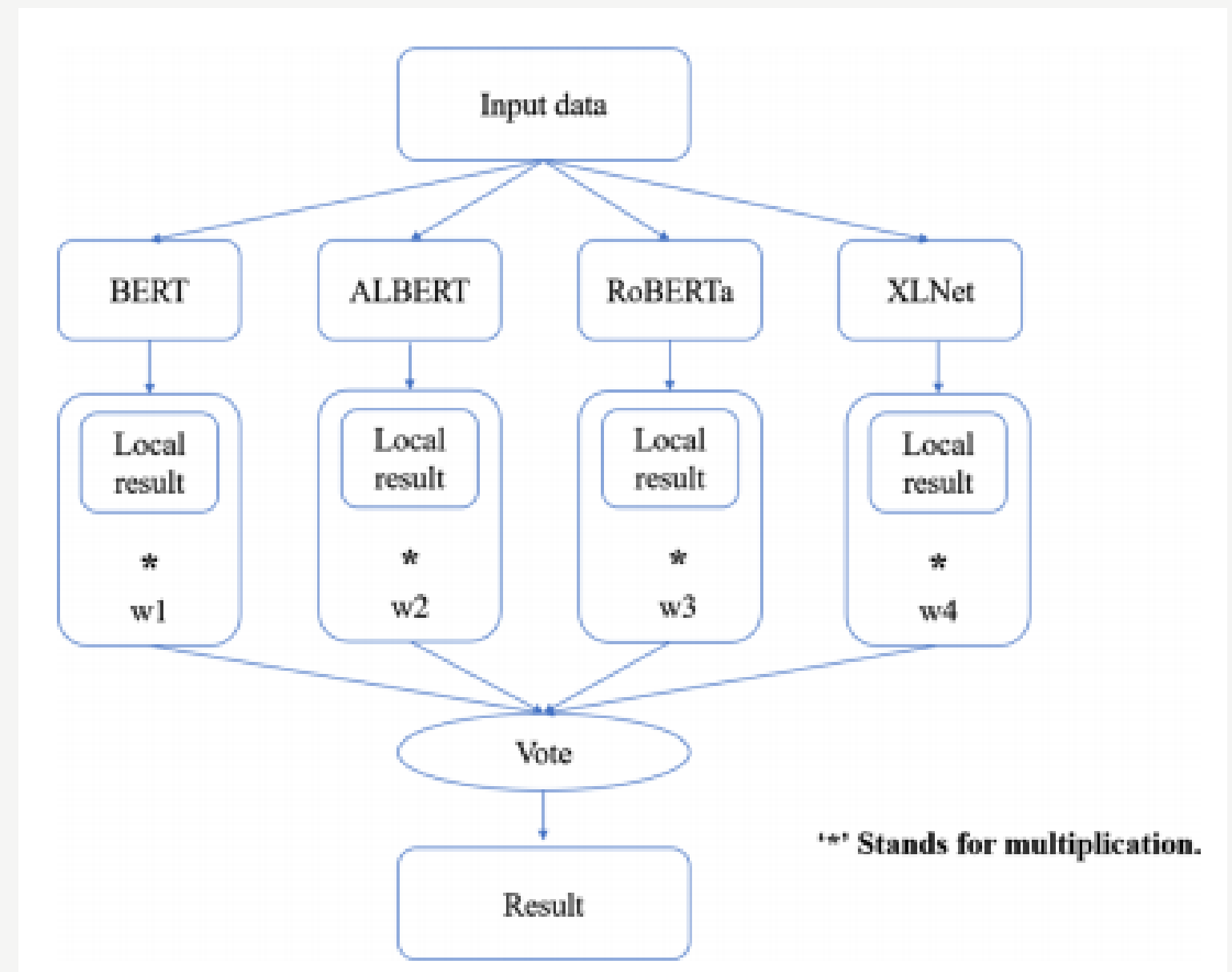
[6]

Performance $\longrightarrow$ Text Generation

- The single classifier with the lowest accuracy is removed from the ensemble model each time to observe the performance .
- The performance of the ensemble learning method is slightly lower than the best single classifier.

Table 20 Results for different combination

Ensemble model	Acc
XLNet+Albert+BERT+GPT2+RoBERTa	0.407
Albert+BERT+GPT2+RoBERTa	0.409
BERT+GPT2+RoBERTa	0.409
GPT2+RoBERTa	0.417
RoBERTa	0.417



[6]

Limitations



I. Computational and Resource Constraints

- Combining LLMs increases the computational complexity, requiring significant resources for inference. [2]
- Latency from multiple forward passes severely constrains usability in real-time or interactive environments (e.g., customer service), leading to response delays that exceed acceptable limits. [1]
- Easy samples, which could be addressed by a single LLM, still waste resources by requiring all LLM candidates to run. [2]
- The resource-intensive nature of ensemble LLMs raises sustainability concerns due to the significant energy consumption and large carbon footprint associated with training and deployment. [1]
- Deep ensembles have significantly more parameters than a single network, posing a challenge for memory-constrained applications. [8]
- Deploying ensembles often requires specialized hardware and distributed systems, which can hinder scalability for organizations with limited resources. [1]
- In output-level LLM ensemble methods, the aggregator receives the outputs of multiple LLMs, which can cause the input sequence length to exceed the limit it can handle, making them unsuitable for complex reasoning tasks that require long, in-depth reasoning. [7]

Limitaions



2. Performance Drawbacks

- Documented failures include incoherent responses or hallucinations arising from unaligned token representations or divergent inference strategies across component models. [1]
- Ensemble LLMs suffer from issues related to coherence and contradiction. Individual models may produce outputs that conflict with one another, leading to inconsistencies in tasks like text generation or summarization. [1]
- If individual component models are homogeneously biased, perhaps due to shared training corpora, the ensemble may amplify rather than reduce these biases. [1]
- Output-level ensemble methods cannot produce a correct answer if all candidate outputs are flawed. [7]

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Thank you

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